Weighted Rank-One Spectral Covering: An $O(d)$ Approximation and Strong Inapproximability

Abstract

We study the following one-sided, unscaled rank-one spectral covering problem. Given vectors $v_1, \dots, v_N \in \mathbb{R}^d$ and positive costs $w_1, \dots, w_N$, find a minimum-cost subset $S \subseteq [N]$ such that

$$\sum_{i \in S} v_i v_i^\top \succeq I_d.$$

The unweighted problem admits a constant-factor approximation through one-sided spectral rounding. For arbitrary weights, we show that the same rounding theorem implies a randomized polynomial-time $O(d)$ approximation. In the opposite direction, we give an approximation-preserving reduction from *Minimum Relevant Variables in a Linear System* (Min-RVLS). The reduction is entirely rank one, uses strictly positive weights taking only two values with polynomial ratio, and has polynomial bit complexity. Consequently, unless $P = NP$, weighted rank-one spectral covering has no constant-factor approximation. More strongly, using the hardness of Amaldi and Kann, for every fixed $\varepsilon > 0$ there is no polynomial-time $2^{(\log d)^{1-\varepsilon}}$ approximation (and likewise with d replaced by the number of input vectors), unless $NP \subseteq DTIME(n^{\mathrm{polylog}\, n})$. In particular, arbitrary weights rule out every polylogarithmic approximation. The hardness construction uses long vectors and therefore does not address the short-vector regime.

1 Introduction

Given vectors $v_1, \dots, v_N \in \mathbb{R}^d$ and positive weights $w_1, \dots, w_N$, consider

$$\min_{S \subseteq [N]} w(S) := \sum_{i \in S} w_i \qquad \text{subject to} \qquad \sum_{i \in S} v_i v_i^\top \succeq I_d. \tag{1}$$

Equivalently, every unit direction $u \in \mathbb{S}^{d-1}$ must receive unit quadratic coverage:

$$\sum_{i \in S} \langle u, v_i \rangle^2 \geq 1.$$

We call (1) Weighted Rank-1 Spectral Cover. The problem is *rank one* because every item contributes a matrix $v_i v_i^\top$, *unscaled* because a selected item is used with coefficient exactly one, and *one-sided* because only a spectral lower bound is required.

The unweighted case $w_i \equiv 1$ is well behaved. One-sided spectral rounding of Lau and Zhou [2] yields an integral cover whose cardinality exceeds a fractional benchmark by only an additive $O(d)$ term. Since every feasible sum of rank-one matrices dominating I_d must contain at least d vectors, this additive term is chargeable to the optimum and gives a constant approximation. For arbitrary weights, however, the same theorem incurs an additive $O(d\, w_{\max})$ cost. A simple cost-thresholding argument converts this into an $O(d)$ approximation.

The main point of this note is that the weighted problem is much harder than the unweighted one. We reduce Min-RVLS—given a feasible integer system $Dx = b$, find a rational solution with minimum support—to Weighted Rank-1 Spectral Cover. The reduction uses cheap rank-one vectors to create an almost-free background geometry. Choosing expensive spectral vectors then becomes exactly the problem of selecting a minimum set of columns of D whose span contains b. This is equivalent to finding a sparsest solution of $Dx = b$.

Amaldi and Kann proved that Min-RVLS with equations is not in APX, and more strongly that for every $\varepsilon > 0$ it is not approximable within $2^{(\log m)^{1-\varepsilon}}$, where m is the number of variables, unless $NP \subseteq DTIME(m^{\mathrm{polylog}\, m})$ [1]. Our reduction preserves the approximation ratio up to a $1 + o(1)$ factor. Thus arbitrary weighted rank-one spectral covering admits no polylogarithmic approximation under the same assumption.

Our results. Let d denote the ambient dimension and N the number of spectral items.

- There is a randomized polynomial-time $O(d)$ approximation for Weighted Rank-1 Spectral Cover (Theorem 3.2).
- Unless $P = NP$, Weighted Rank-1 Spectral Cover has no polynomial-time constant-factor approximation (Corollary 4.6).
- Unless $NP \subseteq DTIME(n^{\mathrm{polylog}\, n})$, for every fixed $\varepsilon > 0$ there is no polynomial-time $2^{(\log d)^{1-\varepsilon}}$ approximation, and the same statement holds with d replaced by N (Theorem 4.5).

The lower bound holds with only two strictly positive integer costs whose ratio is polynomially bounded. The construction does use vectors of potentially exponentially large norm, although all coordinates have polynomial encoding length. Thus the hardness does not extend automatically to instances satisfying $\|v_i\| \le 1$.

2 Preliminaries

For symmetric matrices A, B, we write $A \succeq B$ if $A - B$ is positive semidefinite. For $S \subseteq [N]$, let

$$V(S) := \sum_{i \in S} v_i v_i^\top, \qquad w(S) := \sum_{i \in S} w_i.$$

The optimum of (1) is denoted by $\mathrm{OPT}_{\mathrm{SC}}$.

We use the following one-sided spectral rounding theorem of Lau and Zhou [2, Theorem 1.8]. We state only the part needed here.

Theorem 2.1 (Lau–Zhou). *Let $u_1, \dots, u_N \in \mathbb{R}^d$ and $x \in [0,1]^N$ satisfy*

$$\sum_{i=1}^{N} x_i u_i u_i^\top = I_d.$$

For every $\epsilon \in (0, 1/4)$, there is a randomized polynomial-time algorithm producing $z \in \{0,1\}^N$ such that, with probability $1 - \exp(-\Omega(d))$,

$$\sum_i z_i u_i u_i^\top \succeq I_d.$$

Moreover, for every nonnegative cost vector $c \in \mathbb{R}^N_+$, with the same high-probability guarantee up to adjusting constants,

$$\langle c, z\rangle \le (1+6\epsilon)\langle c, x\rangle + \frac{15d}{\epsilon}\,\|c\|_\infty.$$

We also use the following hardness theorem of Amaldi and Kann [1, Theorem 7].

Theorem 2.2 (Amaldi–Kann)**.** *Consider a feasible integer system $Dx = b$ with m variables, and the problem of finding a rational solution minimizing $\|x\|_0$. This problem is not in APX unless $P = NP$. More strongly, for every fixed $\varepsilon > 0$, it is not approximable in polynomial time within*

$$2^{(\log m)^{1-\varepsilon}}$$

unless $NP \subseteq DTIME(m^{\mathrm{polylog}\, m})$.

3 An $O(d)$ Approximation

The upper bound follows from Theorem 2.1, but two small points are needed. First, the natural fractional solution only satisfies a spectral inequality, whereas Theorem 2.1 assumes equality. This is handled by whitening. Second, the additive term depends on the largest item cost; this is handled by trying all cost thresholds.

For a threshold $W > 0$, let

$$J_W := \{i \in [N] : w_i \le W\}.$$

On J_W, solve the semidefinite relaxation

$$\begin{aligned} \min \quad & \sum_{i\in J_W} w_i x_i \\ \text{s.t.} \quad & \sum_{i\in J_W} x_i v_i v_i^\top \succeq I_d, \\ & 0 \le x_i \le 1 \qquad (i \in J_W). \end{aligned} \tag{2}$$

If it is infeasible, ignore this threshold.

Lemma 3.1 (Whitening)**.** *Let x be feasible for (2), and set*

$$M := \sum_{i\in J_W} x_i v_i v_i^\top \succeq I_d.$$

Define $u_i := M^{-1/2} v_i$. Then

$$\sum_{i\in J_W} x_i u_i u_i^\top = I_d.$$

Furthermore, if $z \in \{0,1\}^{J_W}$ satisfies $\sum_i z_i u_i u_i^\top \succeq I_d$, then

$$\sum_i z_i v_i v_i^\top \succeq M \succeq I_d.$$

Proof. The equality follows by congruence:

$$\sum_i x_i u_i u_i^\top = M^{-1/2}\left(\sum_i x_i v_i v_i^\top\right) M^{-1/2} = I_d.$$

Likewise,

$$\sum_i z_i v_i v_i^\top = M^{1/2} \left(\sum_i z_i u_i u_i^\top \right) M^{1/2} \succeq M.$$

□

Theorem 3.2. *There is a randomized polynomial-time* $122d$*-approximation for* Weighted Rank-1 Spectral Cover*. More generally, any fixed constant choice of* $\epsilon \in (0, 1/4)$ *in Theorem 2.1 gives an* $O(d)$ *approximation.*

Proof. Run the following procedure for every distinct input weight $W \in \{w_1, \ldots, w_N\}$. Solve (2); if feasible, whiten as in Lemma 3.1 and apply Theorem 2.1 with the cost vector $c_i = w_i$ and $\epsilon = 1/8$. Keep the cheapest feasible output over all thresholds and all repetitions. Feasibility of a returned subset can be checked directly. Repeating the randomized rounding polynomially many times boosts the success probability to $1 - N^{-\Omega(1)}$.

Let $S^\star$ be an optimal integral spectral cover, and set

$$W^\star := \max_{i \in S^\star} w_i.$$

At threshold $W^\star$, every item of $S^\star$ is present, so (2) is feasible and its optimum is at most $\mathrm{OPT}_{\mathrm{SC}}$. Also $W^\star \le \mathrm{OPT}_{\mathrm{SC}}$. The Lau–Zhou bound therefore gives, for the successful run at this threshold,

$$\begin{aligned} w(S) &\le \left(1 + \frac{6}{8}\right) \mathrm{OPT}_{\mathrm{SC}} + 15d \cdot 8\, W^\star \\ &\le \frac{7}{4} \mathrm{OPT}_{\mathrm{SC}} + 120d\, \mathrm{OPT}_{\mathrm{SC}} \\ &\le 122d\, \mathrm{OPT}_{\mathrm{SC}}, \end{aligned}$$

where the last inequality uses $d \ge 1$. The algorithm returns a solution no more expensive than this run. □

Remark 3.3 (Unweighted case). If all $w_i = 1$, then every feasible rank-one cover has at least d vectors because a sum of fewer than d rank-one matrices has rank less than d. Hence the additive $O(d)$ term in Theorem 2.1 is already $O(\mathrm{OPT})$, recovering a constant-factor approximation without any thresholding.

4 Hardness via Sparse Linear Systems

We now give a reduction from Min-RVLS. The proof has three ingredients: an exact span characterization, a uniform finite scaling parameter with polynomial encoding length, and approximation-preserving positive weights.

4.1 The source problem

An instance of Min-RVLS consists of a feasible system

$$Dx = b, \qquad D \in \mathbb{Z}^{r \times m}, \quad b \in \mathbb{Z}^r,$$

and asks for a rational solution minimizing $\|x\|_0$. Let

$$k^\star := \min\{\|x\|_0 : x \in \mathbb{Q}^m,\ Dx = b\}.$$

We may assume $b \neq 0$, since $b = 0$ has the trivial zero solution. We may also assume that the rows of D are linearly independent. Indeed, because the system is feasible, retaining a maximal independent set of rows preserves the entire solution set: every discarded row is a linear combination of retained rows, and feasibility guarantees that its right-hand side is the same linear combination of the retained entries of b. Consequently,

$$r \leq m. \tag{3}$$

4.2 The reduction

Let $e := e_{r+1} \in \mathbb{R}^{r+1}$. For the ith column D_i of D, define

$$a_i := \begin{pmatrix} D_i \\ 0 \end{pmatrix} \in \mathbb{Z}^{r+1}, \qquad h_0 := \begin{pmatrix} -b \\ 1 \end{pmatrix} \in \mathbb{Z}^{r+1}.$$

Let

$$H := \max\left\{1, \max_{j,i} |D_{ji}|, \max_{j} |b_j|\right\}.$$

Choose the integer scaling

$$L := 1 + r^{r+1} H^{2r}. \tag{4}$$

The spectral instance has ambient dimension

$$d = r + 1.$$

It contains the following vectors.

(i) For every source variable $i \in [m]$, a *paid* vector

$$u_i := La_i = L\begin{pmatrix} D_i \\ 0 \end{pmatrix}$$

of weight

$$B := (r+1)m^2. \tag{5}$$

(ii) For every $j \in [r]$, a *cheap* coordinate vector $f_j := e_j$ of weight 1.

(iii) One additional cheap vector $h := Lh_0 = L\binom{-b}{1}$ of weight 1.

Thus the construction uses only the two positive integer weights 1 and B. The number of spectral vectors is

$$N = m + r + 1 \leq 2m + 1. \tag{6}$$

The key equivalence is the following.

Lemma 4.1 (Span characterization)**.** *For any $S \subseteq [m]$, the set consisting of all cheap vectors $f_1, \ldots, f_r, h$ together with the paid vectors $\{u_i : i \in S\}$ is a feasible spectral cover if and only if*

$$b \in \operatorname{span}\{D_i : i \in S\}.$$

Proof. Since

$$\sum_{j=1}^{r} f_j f_j^\top = I_{r+1} - ee^\top,$$

feasibility is equivalent to

$$L^2 \left(h_0 h_0^\top + \sum_{i \in S} a_i a_i^\top \right) \succeq ee^\top. \tag{7}$$

Let Z_S be the matrix whose columns are h_0 and the a_i for $i \in S$. Then (7) is $L^2 Z_S Z_S^\top \succeq ee^\top$.

Necessity. If $e \notin \text{range}(Z_S)$, let y be the nonzero orthogonal projection of e onto $\text{range}(Z_S)^\perp$. Then $Z_S^\top y = 0$ while $e^\top y \neq 0$, contradicting $L^2 Z_S Z_S^\top \succeq ee^\top$. Hence $e \in \text{range}(Z_S)$, so

$$e = \alpha_0 h_0 + \sum_{i \in S} \alpha_i a_i.$$

After assigning zero coefficients to unused columns, The last coordinate forces $\alpha_0 = 1$, and the first r coordinates give

$$-b + \sum_{i \in S} \alpha_i D_i = 0.$$

Thus $b \in \text{span}\{D_i : i \in S\}$.

Sufficiency. Suppose $b \in \text{span}\{D_i : i \in S\}$. Choose a linearly independent subset $T \subseteq S$, $|T| = t \leq r$, and coefficients $\alpha \in \mathbb{Q}^T$ such that

$$\sum_{i \in T} \alpha_i D_i = b.$$

Then

$$e = h_0 + \sum_{i \in T} \alpha_i a_i. \tag{8}$$

Write $c = (1, \alpha)$. For every $y \in \mathbb{R}^{r+1}$, Cauchy–Schwarz and (8) imply

$$(e^\top y)^2 = (c^\top Z_T^\top y)^2 \leq \|c\|_2^2 \, y^\top Z_T Z_T^\top y.$$

Therefore

$$Z_T Z_T^\top \succeq \frac{1}{\|c\|_2^2} ee^\top. \tag{9}$$

It remains only to verify that the common scaling L in (4) satisfies $L^2 \geq \|c\|_2^2$. This is proved in Lemma 4.2 below. Then

$$L^2 Z_S Z_S^\top \succeq L^2 Z_T Z_T^\top \succeq ee^\top,$$

which proves feasibility. □

Lemma 4.2 (Polynomial-bit scaling)**.** *For every linearly independent T used in the sufficiency proof above, the coefficients can be chosen so that*

$$\|c\|_2^2 \leq 1 + r^{r+1} H^{2r} \leq L^2.$$

Moreover, the binary encoding length of L is polynomial in the encoding length of (D, b).

Proof. Let $t = |T|$. Since the columns D_T are linearly independent, there are t rows on which they form a nonsingular integer matrix $C \in \mathbb{Z}^{t\times t}$. Restricting the equation $D_T\alpha = b$ to these rows gives $C\alpha = b'$. By Cramer's rule,

$$\alpha_j = \frac{\det C_j}{\det C},$$

where C_j is obtained from C by replacing one column by b'. Since C is nonsingular and integer, $|\det C| \geq 1$. Every column of C_j has Euclidean norm at most $\sqrt{t}\,H$, so Hadamard's inequality gives

$$|\det C_j| \leq t^{t/2}H^t.$$

Hence

$$|\alpha_j| \leq t^{t/2}H^t$$

and

$$\|c\|_2^2 = 1 + \sum_{j=1}^{t} \alpha_j^2 \leq 1 + t^{t+1}H^{2t} \leq 1 + r^{r+1}H^{2r}.$$

By definition, $L = 1 + r^{r+1}H^{2r}$, so certainly $L^2 \geq \|c\|_2^2$. Finally,

$$\log L = O(r\log r + r\log H),$$

which is polynomial in the input encoding length. □

4.3 Approximation preservation

Let $\mathrm{OPT}_{\mathrm{RVLS}} = k^\star$ and let $\mathrm{OPT}_{\mathrm{SC}}$ be the optimum of the constructed spectral instance.

Lemma 4.3 (Optimum comparison)**.** *The constructed instance satisfies*

$$Bk^\star \leq \mathrm{OPT}_{\mathrm{SC}} \leq Bk^\star + (r+1).$$

Equivalently,

$$k^\star \leq \frac{\mathrm{OPT}_{\mathrm{SC}}}{B} \leq k^\star + \frac{1}{m^2}.$$

Proof. For the upper bound, let $S^\star$ be the support of an optimal source solution. By Lemma 4.1, all $r+1$ cheap vectors together with $\{u_i : i \in S^\star\}$ form a spectral cover. Its cost is $Bk^\star + (r+1)$.

For the lower bound, consider an arbitrary feasible spectral solution T, and let

$$S := \{i \in [m] : u_i \in T\}$$

be its paid indices. Add all cheap vectors that T may have omitted. Spectral feasibility is monotone under adding PSD summands, so the augmented set remains feasible. Lemma 4.1 implies $b \in \mathrm{span}\{D_i : i \in S\}$; hence there is a source solution supported on S and therefore $|S| \geq k^\star$. Since every paid vector has weight B,

$$w(T) \geq B|S| \geq Bk^\star.$$

The equivalent form follows from $B = (r+1)m^2$. □

Lemma 4.4 (Solution extraction)**.** *From any feasible spectral solution T of cost C, one can compute in polynomial time a feasible source solution x satisfying*

$$\|x\|_0 \leq \frac{C}{B}.$$

Proof. Take the paid support S of T. As in Lemma 4.3, adding all cheap vectors and applying Lemma 4.1 gives $b \in \operatorname{span}(D_S)$. Solve $D_S x_S = b$ by Gaussian elimination and set all other coordinates to zero. Since $C \geq B|S|$, the support size is at most C/B. □

Theorem 4.5 (Strong inapproximability)**.** *Unless $NP \subseteq DTIME(n^{\operatorname{polylog} n})$, for every fixed $\varepsilon > 0$ there is no polynomial-time $2^{(\log d)^{1-\varepsilon}}$ approximation for* Weighted Rank-1 Spectral Cover*. The same statement holds with d replaced by N, the number of spectral vectors.*

The hardness holds even when all matrices are rank one and the weights take only two positive integer values 1 and B, with B polynomially bounded in the source instance size.

Proof. Suppose, toward contradiction, that for some fixed $\varepsilon > 0$ there is a polynomial-time spectral covering algorithm with approximation ratio

$$\rho(d) = 2^{(\log d)^{1-\varepsilon}}.$$

Apply it to the reduction above. By Lemmas 4.3 and 4.4, the resulting source solution has support at most

$$\rho(d)\left(k^\star + \frac{1}{m^2}\right) \leq \rho(d)\left(1 + \frac{1}{m^2}\right)k^\star,$$

because $k^\star \geq 1$. By (3), $d = r + 1 \leq m + 1$. Hence, for all sufficiently large m,

$$\rho(d)\left(1 + \frac{1}{m^2}\right) \leq 2^{(\log m)^{1-\varepsilon/2}}.$$

This contradicts Theorem 2.2, applied with $\varepsilon/2$.

For a ratio parameterized by the number of spectral vectors, use (6): $N \leq 2m + 1$. Again, after replacing ε by (say) $\varepsilon/2$, the constant-factor change inside the logarithm is absorbed for sufficiently large m, contradicting Theorem 2.2.

Finally, $B = (r + 1)m^2 \leq (m + 1)m^2$ is polynomially bounded, and Lemma 4.2 shows that all vector coordinates have polynomial encoding length. Thus the reduction is polynomial time in the standard bit model. □

Corollary 4.6. *Unless $P = NP$,* Weighted Rank-1 Spectral Cover *has no polynomial-time constant-factor approximation.*

Proof. The same reduction, together with the fact that Min-RVLS with equations is not in APX unless $P = NP$ [1], transfers any constant-factor approximation for Weighted Rank-1 Spectral Cover to a constant-factor approximation for Min-RVLS by Lemmas 4.3 and 4.4. □

Corollary 4.7 (Polylogarithmic hardness)**.** *Under the assumption $NP \not\subseteq DTIME(n^{\operatorname{polylog} n})$,* Weighted Rank-1 Spectral Cover *admits no $(\log d)^C$ approximation for any fixed constant C, and likewise no $(\log N)^C$ approximation.*

Proof. For every fixed C and every fixed $\varepsilon \in (0, 1)$,

$$(\log n)^C = 2^{C \log\log n} = o\left(2^{(\log n)^{1-\varepsilon}}\right).$$

Thus a polylogarithmic approximation would be stronger than the factor excluded by Theorem 4.5 for all sufficiently large instances. □

5 Discussion and Open Directions

The two theorems leave a substantial approximation gap. The upper bound is $O(d)$, while the lower bound rules out every polylogarithmic approximation but is still only subpolynomial:

$$2^{(\log d)^{1-\varepsilon}} = d^{1/(\log d)^{\varepsilon}} = d^{o(1)}.$$

Thus the present results do *not* give an essentially tight characterization. Natural next questions include whether one can obtain a polynomial improvement over the $O(d)$ upper bound, or strengthen the hardness to $d^{\Omega(1)}$.

The reduction also identifies an important boundary. It relies on a large scaling L and therefore produces long vectors. It does not rule out substantially better approximations in the short-vector regime $\|v_i\| \leq 1$. This is consistent with the role of bounded vector norms in matrix concentration and spectral rounding. A separate study of weighted short-vector spectral covering may therefore have a very different approximability landscape.

Finally, the hardness construction explains why arbitrary weights change the problem qualitatively. The cheap vectors create an almost-free spectral background, and the expensive vectors are then used only to resolve a sparse linear-representation question. This mechanism is unavailable in the purely unweighted problem, where the background itself costs $\Theta(d)$ and the additive $O(d)$ loss of spectral rounding can be charged to the optimum.

References

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